RPMGLOBAL

AMT –Maintain Centrelines

Training Manual

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RPM SOFTWARE – ASSET MANAGEMENT

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Preface

The training workshops are developed to supplement instructor-led training. The guide is constructed in a way that teaches students the key functionality of the application through practical, hands-on workshops. Each workshop details tasks that the student is required to complete. The workshops build on each other to help ensure a comprehensive understanding of the application is achieved at the completion of the course. Space is provided at the end of each workshop for learning checks and student notes.

Symbol Library



Note

A note for participants about the task.



Important Reminder

An important reminder for the participant about the task.



Learning Check

Check your understanding on the workshop and ask any related questions.

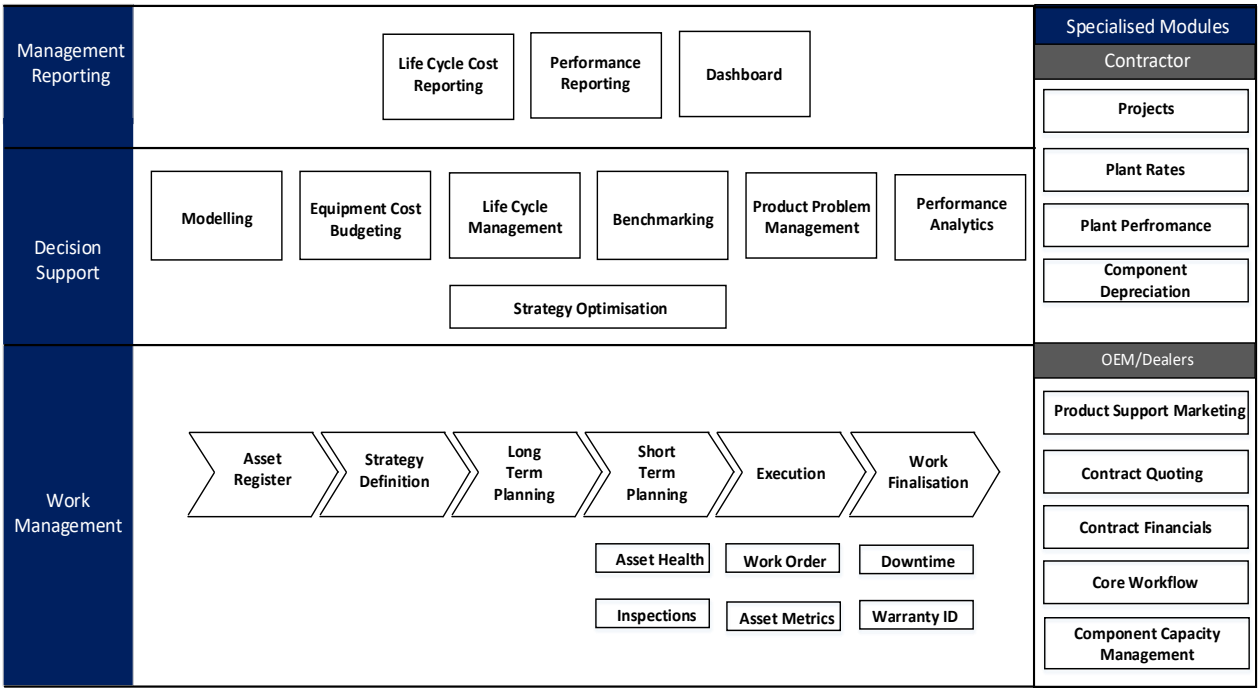
Course Outline

Pre-requisites	Getting Around AMT, AMT Key Concepts.
Target Audience	Asset group
Nominal Duration	
Course Objectives	
Learning Outcomes	At the end of the course, participants will have learned to: • Basic setup. o Review centreline-related system tables. o Create centreline hierarchies (branch/site/fleet). o Manually register a centreline. o Set the term for an asset. o Copy/import/create strategy tasks. • Usage profiles. o Set up usage profiles. o Link assets to usage profiles. • Create Centreline Strategy Tasks o Copy a Projection o Import a Projection o Create Strategy Tasks for the Centreline's Projection • Link Standard Jobs to Centreline Strategy Tasks • Create Unassigned Tasks

	• Link Component Structures • Analyse Centrelines • Validate Life Cycle Costs • Validate and Publish
Assessment Strategy	The completion of workshop tasks.
AMT version	8.26

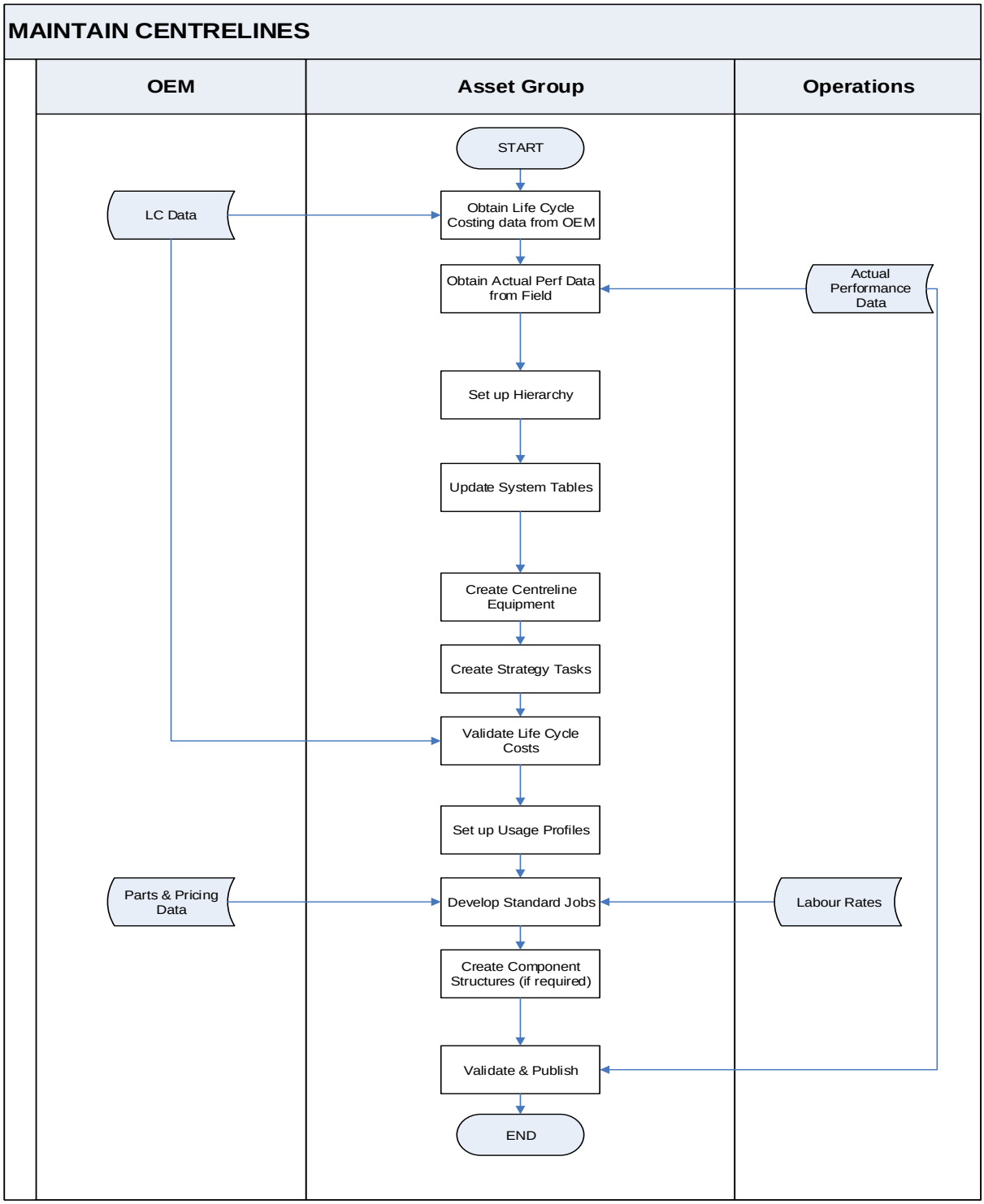
Introduction to AMT

AMT is a comprehensive asset management system that consists of integrated modules as depicted below.



Processes

Process Diagram



Process Inputs

The following inputs are required for this process:

1. Estimated component life cycles per model of equipment.
2. Estimated costs for each replacement of components, fluids, and consumables.
3. Characteristics of the different applications in which the assets or equipment would operate.

Process Outputs

The following outputs can be expected from this process:

1. Populated templates that support the assets/equipment used by the company.
2. Templates that have reliable life cycles for components.
3. Templates that reflect today's prices or costing.
4. Templates that are sensitive to the needs of the site or project (for example, cater for different usage and application profiles).

Training Workshops

1. Obtain Life Cycle Costings from OEM

Overview

This process is an input to the Maintaining Centrelines process.

Relevant life cycle costing models should be obtained from OEMs, which include the following details:

1. Life cycle of the equipment.
2. Estimated component lives and replacement costs.
3. Estimated labour requirements for repairs.
4. Service regime, estimated servicing costs, and labour requirements.
5. Estimated purchase price.
6. Optional configurations (such as sound suppression, fire suppression).
7. Consumable and fluid requirements.

2. Obtain Actual Performance Data from the Field

Overview

This process is an input to the Maintaining Centrelines process.

Equipment that is currently operating in the field can provide useful data to establish and verify the running costs of equipment in specific applications. Try to gather this data from the field:

1. Realistic equipment life cycles.
2. Actual component life, replacement cost and labour data.
3. Actual servicing costs and labour required.
4. Actual consumable and fluid requirements.

3. Update System Tables

Procedure – Update System Tables

To create a centreline, you need basic system table data such as model and manufacturer details. Such entries may need to be created in AMT's system tables to support the centreline.

In this step, the existing system tables are updated. This update can involve changes to equipment-related maintenance and usages. Review the list of system tables below in case additional options are needed.

Typically, the AMT Administrator has access to make these changes, so a formal request to that person may be required.

Table Type	Table name	Description
Equipment	Equipment Class	High-level grouping of equipment (such as Mobile, Fixed Plant).
Equipment	Equipment Group	A subsection of Equipment Class (such as Off Highway Truck).
Equipment	Manufacturer	The company that builds the equipment (such as Boeing).
Equipment	Model	The model of equipment (such as A380, RH400, 793C).
Usages & Measurements	Unit of Measure	Life cycle measurement unit (such as engine hours, fuel).
Maintenance	Cost Type	Grouping of different categories of cost (such as repair & maintenance, finance, operator).
Other	Component Code	Specific cost items within each cost type (such as engine, transmission, undercarriage, fuel, tyres, insurance, registration).
Maintenance	Modifier Code	Position code (such as front, rear, top, bottom, front left, front right).
Maintenance	Job Code	Describes the action performed (such as remove and install, recondition, service, weld).

Learning Check



Check your understanding of the workshop and ask any related questions.

Notes

4. Set Up Equipment Hierarchy

Overview

Every asset in AMT is linked to a fleet, which belongs to a site, which belongs to a branch. This applies to centrelines as well. When creating centrelines, we must ensure that the relevant fleet, site or branch that the centreline belongs to exists in AMT.

Using an incorrect equipment hierarchy for a centreline can have implications. Let's assume you create a centreline branch or place centrelines under the branch that reflects HQ because centrelines are standard and not site-specific. In this case:

- If the centreline's strategy tasks are linked to standard jobs, the hierarchy of the centreline will have implications on how parts are priced (fleet) and which labour rates are used (branch). The parts pricing applied will be based on the centreline fleet's pricing group. The labour rate applied will be based on the combination of the branch and the labour activities used in the standard job linked to the strategy tasks within the centreline.
- If a centreline's strategy tasks are not linked to a standard job, a user can enter high-level cost for parts, labour or miscellaneous manually.

It is not unusual for centreline tasks to have manually entered costs and labour hours (as opposed to a detailed standard job), because a centreline, by definition, is not specific to a location (it is a template).

Procedure – Set Up Equipment Hierarchy

Validate the equipment hierarchy to confirm that the desired hierarchy exists. If not, you need to edit or create the hierarchy to match the requirements of the desired centreline.

Perform the following steps to validate, edit or create the centreline equipment hierarchy:

1. To validate that a branch exists, use the **Search** field to search **Branch Administration**. Double-click **Branch Administration** under the **Centrelines** category.



*Alternatively, from the **My Functions** hierarchy, select **Centrelines** > **Administration** > **Branch Administration**.*



*By default, AMT groups the functions for all Centrelines in a separate category called **Centrelines**. You can directly access this category of functions from the **My Functions** hierarchy to validate if the desired hierarchy exists.*
*In the default hierarchy, AMT has a branch and site by the name of **Centrelines**. You may choose to create new as per your requirements.*

2. Click **Apply Filter** on the **Branch Administration** dialog.

3. Check that the branch exists for the centreline. If so, click **Close**.



*• AMT displays a list of all branches because the **Branch Administration** function is the same function as under modelling and in the top hierarchy. Even though **Branch Administration** is in the centreline category, it is not specific to it.*

• However, there are some centreline-specific functions that have a "- Centreline" in their name.

4. If the branch does not exist, click **Add**.
5. Enter the **Branch** name in the **Add Branch** dialog and click **Save**.



If the branch created in AMT does not exist in the ERP, the labour rates need to be maintained manually in AMT. However, as stated previously, if standard jobs are not used to support centrelines, this consideration is not applicable.

6. To validate that the site you want exists, search **Site Administration** using the **Search** field.
7. Double-click **Site Administration** under the **Centrelines** category.



*Alternatively, from the **My Functions** hierarchy select **Centrelines** > **Administration** > **Site Administration**.*

8. Click **Apply Filter** on the **Site Administration** dialog to view all the sites under the centrelines category.
9. Validate the site exists for the centreline. If so, click **Close**.
10. If the site does not exist, click **Add**.
11. On the **Add Site** dialog, perform the following actions:
    i. Enter the **Site** name and the associated **Branch** name as required.
    ii. Set the **Currency** ("AUD" in the example).
    iii. Click **Save**.
12. To validate that a **Fleet** you want exists, use the **Search** field to search for **Fleet Administration**.
13. Double-click **Fleet Administration** under the **Centrelines** category.



*Alternatively, from the **My Functions** hierarchy select **Centrelines** > **Administration** > **Fleet Administration**.*

14. Click **Apply Filter** on the **Fleet Administration** dialog to view all the fleets in AMT and validate if the desired one exists.

15. Check that the fleet exists for the centreline. If so, click **Close**.

16. If the fleet is missing, you need to create it. To create a fleet, click **Add**.

17. On the **Add Fleet** dialog, perform the following actions:

    i. Select **Fleet**.
    ii. Select **Site**.
    iii. Select **Pricing Group**.
    iv. Set the **Service Interval:** example 250 Hrs (mandatory in AMT).
    v. Set the **Tolerance:** example 10% (mandatory in AMT).
    vi. Click **Save**.



If standard jobs are used in the centreline, the parts pricing applied will be based on the centreline fleet's pricing group. However, as stated previously, if standard jobs are not used in centrelines, this consideration is not applicable.

*The **Service Accuracy Details** section has zero impact on LCC. However, the fields in this section are mandatory to fill in AMT. These fields are only relevant for the work management functionality of AMT.*

Learning Check



Check your understanding of the workshop and ask any related questions.

Notes

5. Create Centreline Equipment

Overview

After you've made the required changes to the equipment hierarchy and system tables, it's time to create the centreline.

Along with the new centreline, we also define its strategy tasks, either in AMT or copy strategy tasks from a real equipment. Further adjustments can be made to the centreline, if required.

There are two ways to ensure that other users of AMT do not use a work-in-progress centreline:

- Use different site or fleet combinations.
- Name the centreline equipment to show it's in progress.

Naming the centrelines to indicate their status – published or work in progress – is the simplest way. For example, you could include the letter **P** in the equipment name to indicate it is published and include the letter **D** to indicate it is draft.

There are two ways to create centreline equipment in AMT:

1. Create centreline from an existing equipment.
2. Create a new blank centreline.

Procedure – Create Centreline from an existing Equipment

Perform the following steps to create a centreline from an existing equipment:

1. Search **Copy and Create Equipment - Centreline** using the **Search** field and double-click **Copy and Create Equipment - Centreline**.



*Alternatively, from the **My Functions** hierarchy select **Centrelines** > **Administration** > **Copy and Create Equipment - Centreline**.*

2. On the **Copy and Create Equipment** dialog, perform the following actions:
    - **Source:** Enter the name of the equipment to copy and its projection.
    - **Destination:** Enter the destination fleet name and ensure that you select **Centreline** as the **Equipment Type**.
    - **Generic Details:** Enter the **End Usage** and **Pricing Date**.
        - **Pricing Date** is the date from which the costs ("prices") of a strategy task are valid. AMT has an escalation logic, which uses the **Pricing Date**.
    - **Equipment Details:** Enter the **Equipment Name** and **Start Date**.
    - The **Equipment Name** must be unique in AMT. We recommend that the name clearly indicates that it is a centreline. You could do this by using a suitable prefix in the name, such as **C** or **C-**.
    - Click **Process**.

3. On the **Copy Projection** dialog, select the **Leave the pricing as is** option and click **OK** to create the centreline projection.



This process copies the source equipment's term, utilisation and strategy tasks into the centreline equipment.

Procedure – Create a new blank Centreline

Perform the following steps to create a new blank centreline:

1. Search **Create Blank Equipment - Centreline** using the **Search** field and double-click **Create Blank Equipment - Centreline**.



*Alternatively, from the **My Functions** hierarchy select **Centrelines** > **Administration** > **Create Blank Equipment – Centreline.***

2. On the **Add Equipment** dialog, perform the following actions:
    - **Equipment Name**: Enter a name for the equipment.
    - Select a **Fleet**.
    - Select a **Manufacturer**.
    - Select a **Model**.
    - Select a **Utilisation:** Choose whether the utilisation is **Simple** or **Advanced**.
        - **Simple**: Only one unit of measure for the asset, for example, dates.
        - **Advanced**: More than one unit of measure.
3. To set utilisation assumptions, click the **Term** tab and perform the following actions:
    - Select a **Primary UOM**.
    - Enter the **Annual Utilisation**.
    - Enter the **End Usage**.
4. To set cost details, click the **Classifications** tab. On the **Default Cost Bearer** field, select the relevant **Default Cost Bearer**.



*Options available in the **Classification** fields come from AMT system tables. Contact your system administrator to add additional options to these fields, if required.*

5. To save the changes:
    - Click the **Term** tab.
    - Click **Refresh**.
    - Click **Save**.
6. The Centreline is created.

Learning Check



Check your understanding of the workshop and ask any related questions.

Notes

6. Define Usage Assumptions

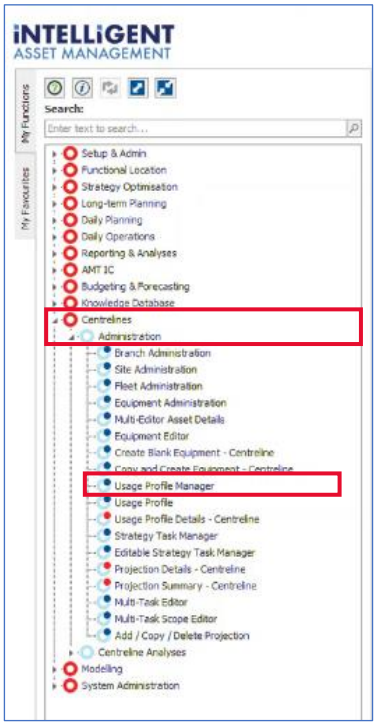
Overview

When you create the centreline, you define relevant usage assumptions for it. In addition, some strategy tasks should be forecasted using measurements other than the equipment's default unit of measure. For example, the undercarriage wear of an excavator is sometimes better measured by travel hours rather than by machine hours (even though it may be common to use machine hours).

Procedure – Define Usage Assumptions

Perform the following steps to define usage assumptions for a new centreline:

1. To create a usage profile, use the **Search** field to search for **Usage Profile Manager** and double-click **Usage Profile Manager**.



*Alternatively, from the **My Functions** hierarchy select **Centrelines** > **Administration** > **Usage Profile Manager.***

2. The **Usage Profile Manager** dialog displays. Click **Add** .

3. On the **Add Usage Profile** dialog, perform the following actions:

   - **Usage Profile:** Enter the relevant usage profile name.
   - **Unit of Measure (UOMs):** Click **Add**.

*UOMs can be a child of **Calendar Year** or can be a child of another UOM. For example, fuel burn could be child of machine hours so that for every 1 machine hour "x" amount of fuel is consumed.*

4. On the **Unit of Measure** dialog, perform the following actions:
    - **Usage per parent UOM:** Enter an appropriate value.
    - **Unit of Measure:** Select the required UOM.
    - Click **Save**.

*New units of measure can be added to AMT using the **Unit of Measure** system table.*

5. Click **Save** on the **Add Usage Profile** dialog.
6. Click **Yes** on the warning message to proceed.



7. Access the **Edit Equipment** dialog by following these steps:
    - Click **Centrelines** on the AMT home page.
    - Click **Apply Filter** to display all centrelines.
    - Click **OK** on the **Equipment Selector** dialog.
    - Select the **Centrelines** option from the **Centreline: Equipment** section.
    - Double-click the newly created centreline name.
8. Select **Advanced Utilisation** in the **Utilisation** section.
9. To add the usage profile, click the **Term** tab and perform the following actions:
    - **Usage Profile:** Enter the usage profile.
    - Click **Refresh**.
9. Click **Yes** on the message to proceed.
10. To view or edit the usage profile, click **Quick Links** > **Usage Profile**.
11. The **Edit Usage Profile** dialog displays. Make the required changes to the usage profile.
12. Click **Save**, and you will return to the **Edit Equipment** dialog for the new centreline.

Learning Check



Check your understanding of the workshop and ask any related questions.

Notes

7. Create Strategy Tasks

Overview

Now that the centreline is registered, its costs have to be built up using data gathered from OEMs and the field as inputs. The costs are recorded as strategy tasks in the centreline's projection.

The centreline has a projection type called centreline, which has the following characteristics:

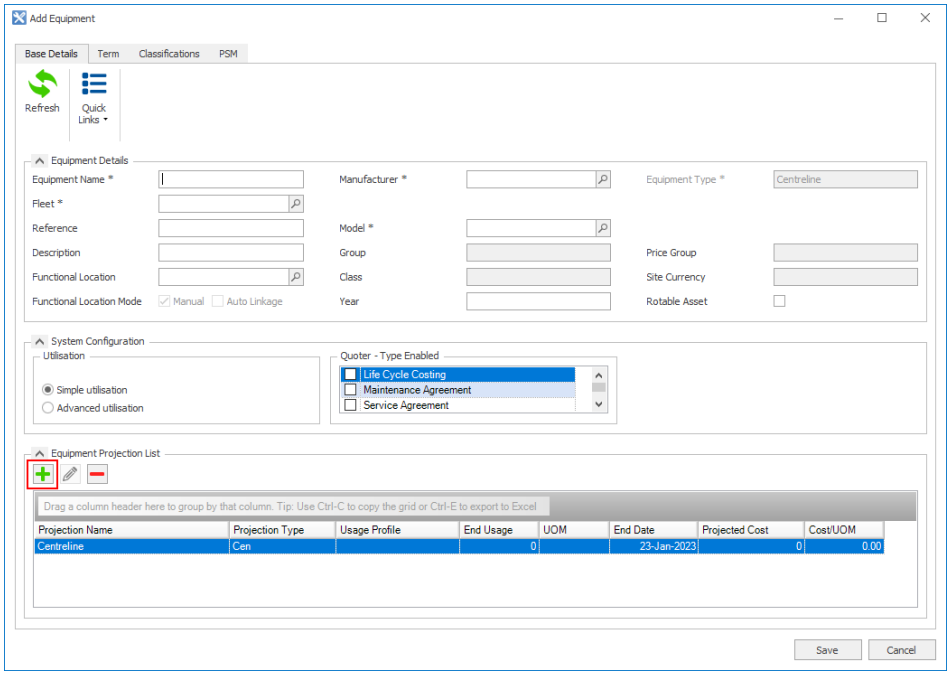
- Always assumed to be current in terms of costing and accurate component life cycles.
- Cannot have a start date in the past – AMT updates the start date at the start of each month so scheduled tasks can never become overdue.
- Strategy tasks can be linked to standard jobs.

If a real equipment currently registered in AMT has a suitable projection on which to base the centreline, it can be copied to the centreline's projection. Otherwise, import the projection (from an external system or load sheet), or create the strategy tasks for the centreline's projection as new.

Procedure – Copy a Projection

Perform the following steps to copy a projection to a centreline:

1. On the **Equipment Projection List** panel, click **Add**.



2. Set the **Projection Type** to be **Centreline**.
3. Select **Use Existing**.
4. Select **Equipment Type**, **Equipment** and **Projection**.
5. Click **Process** to copy the selected equipment's projection into the centreline.

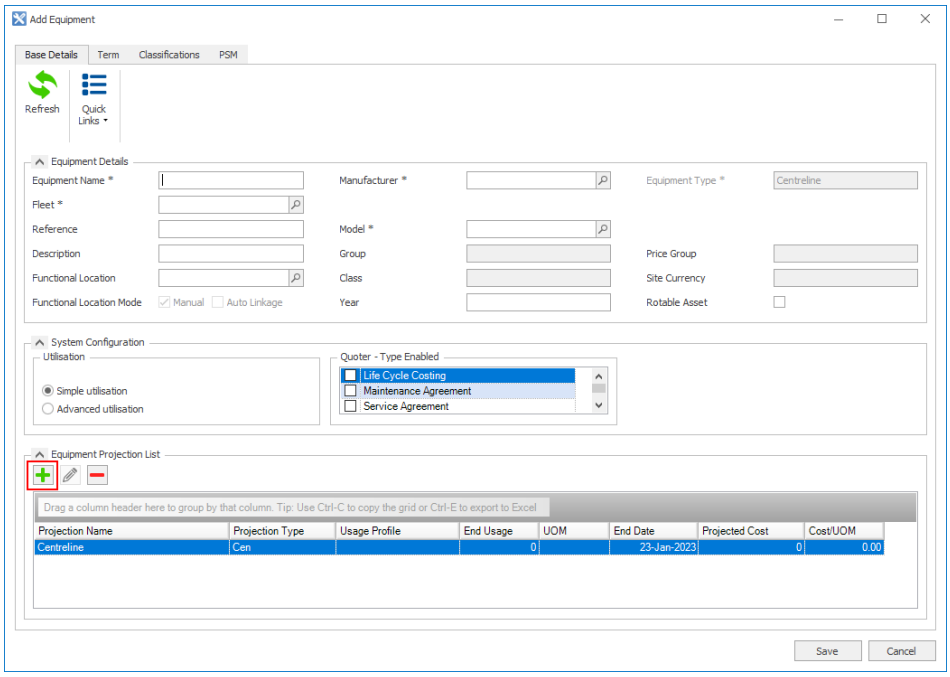
This process will delete the centreline's existing projection and replace it with the selected equipment's projection. Cost sources from the source projection will be maintained.

Depreciable asset structures will not be copied.

Procedure – Import a Projection

Perform the following steps to import a projection to a centreline:

1. On the **Equipment Projection List** panel, click **Add**.



2. Set the **Projection Type** to **Centreline**.
3. Select **Import Projection**.
4. Click the **ellipsis**. Navigate to and select the file to import.

The import file must be a CSV file of a specific structure. Contact your AMT system administrator for more information.

5. Select the **Equipment** and its **Projection** from the file.

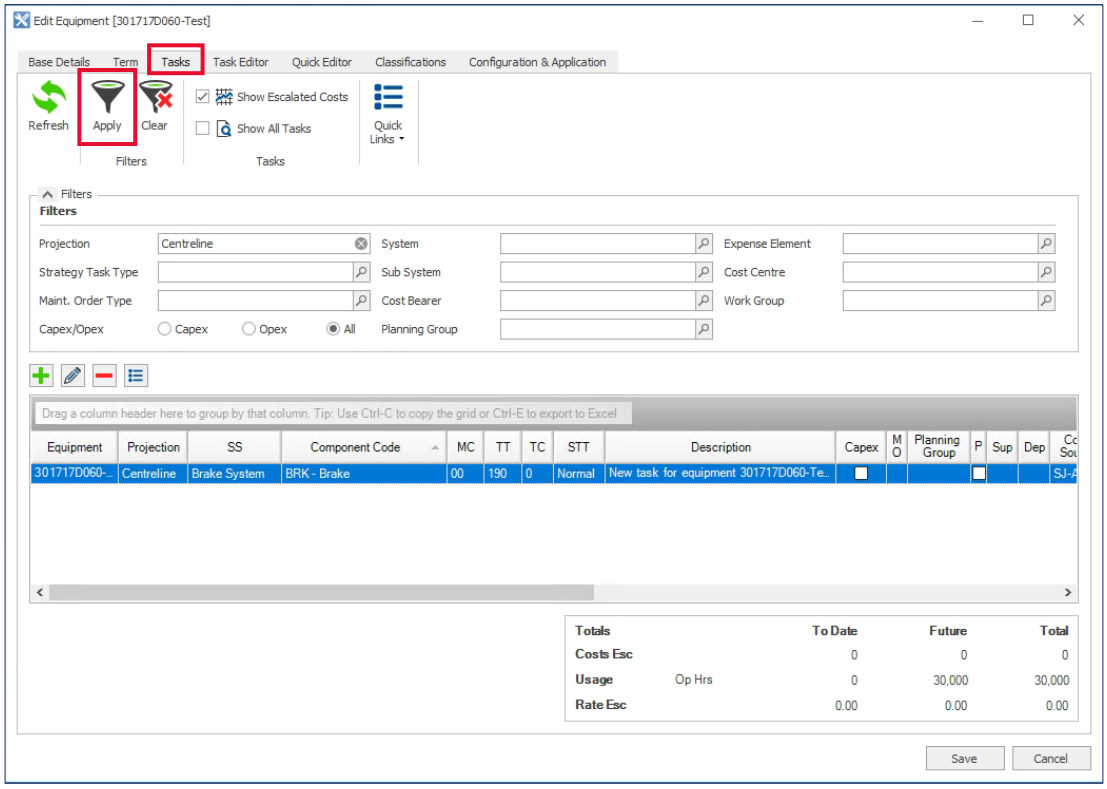
*If errors display when you click the **Equipment** filter, it means that the import file is not formatted correctly, resulting in AMT not being able to locate specific equipment in the file.*

6. Click **Process** to import the selected file into the centreline's projection.

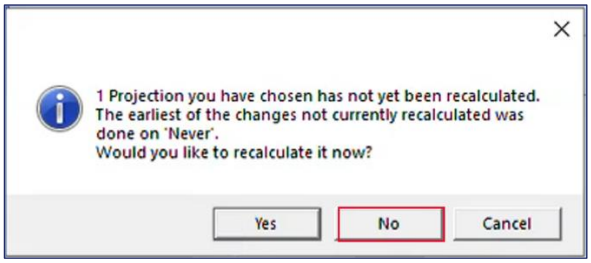
Procedure – Create Strategy Tasks for the Centreline's Projection

Perform the following steps to add strategy tasks to a centreline:

1. On the **Edit Equipment** dialog, click the **Tasks** tab for the new centreline. Then click **Apply Filter**.



2. On the recalculation notification box, click **No** (because recalculation is not required, since the centreline does not contain any strategy tasks).



Recalculation would be required if the centreline was copied from another equipment.

3. Click **Add** to add a new task to the centreline you created.

4. On the **Add Task** dialog, perform the following actions:

   - Select **Normal Task**.
   - Click **OK**.

***Normal Task** is used for specific cost items (such as motor, cylinder, service, inspection).*

***Unassigned Task** is used to set up provisions for costs within a specific subsystem.*

***Other Operating Cost Task** is used to set up other operating costs.*

5. The **Add Strategy Task** (linked to the centreline) dialog displays. By default, the **Base Details** tab displays.

   To define the strategy task, perform these actions on the **Base Details** tab:

   - Enter **Description** for the task.
   - Select a **Component Code**.
   - Select the **Task Type**.
   - Select the **Task Mode**.

*If you create an **Unassigned Task**, the task codes are predetermined and cannot be changed.*

*If you wish to set up group strategy tasks into a suppression group, **Planning Task** mode must be selected.*

*In the **Planning Task** mode, the **Planning Lead Time** field is a mandatory field to fill in AMT. However, if AMT is not being used for Work Order Management, this field has no actual impact.*

6. Click the **Scheduling** tab and perform the following actions:

   - Select **Scheduling Method**.
   - Enter Task **First Occurrence:** The first occurrence value.
   - Enter Task **Frequency**
   - If you selected an unassigned task, **First Occurrence** is not applicable.

- If you selected **Planning Task mode**, **Advanced Scheduling Logic** options display:

    - **Parent Suppression** is the most used suppression logic.
    - In the **Suppression Grouping** field, select a group based on the contents of the Scheduling Group system table in AMT.
    - In the **Suppression Priority** field, set the relevant priority. A priority of **1** is the top of the tree and suppresses children **2** under it, which suppresses **3** under it, and so on.
    - All of this depends on the **First Occurrence** and **Frequency** parameters of each strategy task linked to the same scheduling group within the same equipment.
    - A **Scheduling Group** itself can be used by several equipment, but the suppression schedule is limited to the strategy tasks linked to a scheduling group in single projection linked to a single equipment. For example, a scheduling group could be “PM Services”, which several equipment use, but each of the equipment’s strategy tasks calculate their own suppression schedule.
        - **Tolerance:** For example, you would not change the turbo on the engine and then, 48 hours later, replace the entire engine, which includes a brand-new turbo.
        - The **Tolerance** value is an attribute of the selected suppression grouping. In the example shown, suppression group “B” has a tolerance of “48”.
        - Users can change tolerances.



Refer to the AMT Scheduling Methods training manual to learn more about AMT’s Advanced Scheduling Logic

7. Click the **Job Details** tab and perform the following actions:

    - Select **Cost Source.** A strategy task can have multiple jobs linked to it. The cost source options for a single job are:

        - **Manual**: you can manually enter the costs.
        - **Standard Job**: costs are sourced from standard jobs in AMT. Refer to the Standard Job training document for more information.
        - **Cost Allocation**: you can define multiple cost elements across different cost centres and expense elements for a single job.
        - **Consumables**: you can enter quantity, top up and unit price to calculate the parts cost element of the job.
    - Enter **Labour Hours**.
    - Enter the amount of **Labour Cost**.
    - Select **Labour Expense Elements.**
    - Click **Save**.

There is no link between labour hours or labour activity and the calculation of labour cost.

The pricing date reflects the date from when the costs entered are valid. It is from this date that escalation will be applied.

The duration is the amount of time required to complete the job.

The level of detail defined depends on the purpose of the centreline.

Centrelines are not used to set up equipment in the ERP. So, the level of detail required is normally less than the detail you will see within a real equipment imported from the ERP.

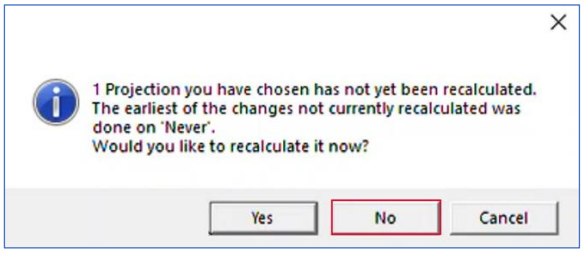
Unless required for analysis, you do not need to select a labour activity.

You can select an expense element, but it is not required.

You can link a centreline strategy task to a standard job, but it is not required.

8. Click **Apply Filter**.

9. Every time you create a new task, you're asked if you want to recalculate. Click **No,**as calculation is not required at this stage.



The new strategy task is listed in the dialog.

Learning Check



Check your understanding of the workshop and ask any related questions.

Notes

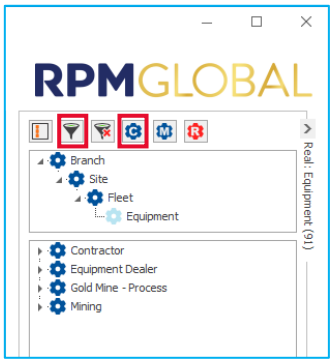
8. Create AMT Unassigned Tasks

Procedure – Create AMT Unassigned Tasks

In this step, we create unassigned tasks relevant to the centreline.

Perform the following steps to define unassigned tasks for a new centreline:

1. To select a centreline, on the AMT home page:
    - Click the **Centrelines** icon.
    - Click **Filter** to display all centrelines.



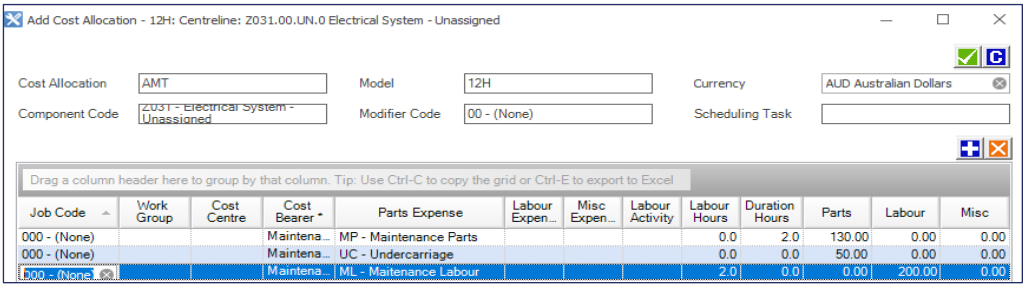
    The **Equipment Selector** dialog displays.
2. Click **OK** to display all centreline equipment.
3. To view the newly created centreline, select the **Centrelines** option from the **Centreline Equipment** section.
4. Double-click the centreline name to view and edit its details. The **Edit Equipment** dialog displays.
5. Click the **Tasks** tab.
6. In the **Tasks** tab, click **Add**.
7. The **Add Task** dialog displays. To create an **Unassigned Task**:
    - Select **Unassigned Task**.
    - Click **Search** in the **Subsystem** field. Double-click to select the appropriate subsystem.
    - Click **OK**.

    The **Edit Strategy Task** dialog displays.
8. To set the task schedule, click the **Scheduling** tab.
9. In the **Scheduling** tab, perform the following actions:
    - Select a **Scheduling Method.**
    - Enter the **Frequency**.
10. To set the task's cost and other job details, click the **Job Details** tab.
11. On the **Job Details** tab, perform the following actions:
    - Enter **Parts**.
    - Enter **Labour** cost.
    - **Enter the Labour Hours**.
    - Click **Save**.

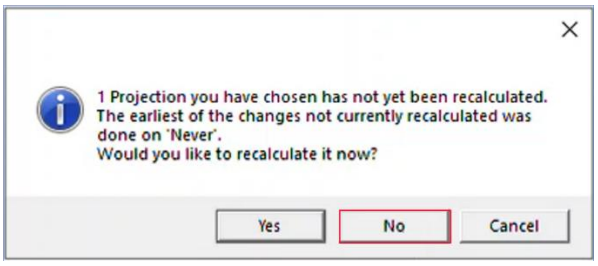
To define the costs for an unassigned task with multiple expense elements:

- Set the **Cost Source** as **Cost Allocation**.
- Click **Add**.

  The **Add Cost Allocation** dialog displays.
- Add rows within the cost allocation by clicking **Add** to represent the breakdown of costs per expense element.
- Click **Save and Close**.



11. In the **Edit Strategy Task** dialog, click **Refresh**.
12. To view or edit the details, click **Quick Links** > **Cost Allocation**. Otherwise, click **Save**.
13. Click **Apply Filter** in the **Edit Equipment** dialog.
14. The recalculation notification displays. Click **No**.



15. The strategy task and the unassigned task associated with the centreline display. To view the recalculation due to the addition of the unassigned strategy tasks, click **Apply Filter**.
16. The recalculation notification displays. Click **Yes** and wait for the recalculation to conclude.
17. The **Future Costs Esc**, **Total Costs Esc** and other columns are now populated. Click **Save**.

Learning Check



Check your understanding of the workshop and ask any related questions.

Notes

9. Link Component Structures

Overview

If the CBM (Condition Based Maintenance) module is installed, you may want to create component structures to refine strategy task cost projections further. A component structure defines the failure modes for a specific component (strategy task) for a specific model of equipment. Component structures can only be created if there is sufficient knowledge of the component's failure modes.

Linking a strategy task to a component structure does not override the task's existing scheduling parameters or costs. Rather, the component structure is recalculated using the task's original values using ratios.

Ultimately, linking to a component structure results in:

- A more detailed definition of the maintenance strategy for this task, which can be viewed by clicking **Quick**.
- The ability to create a repair reserve.
- Define CBM measurements for each failure mode, which will be useful when the centreline is applied to a real equipment/asset.

Refer to the CBM training document for the steps to create and link component structures to strategy tasks.

10. Analyse Centrelines

Overview

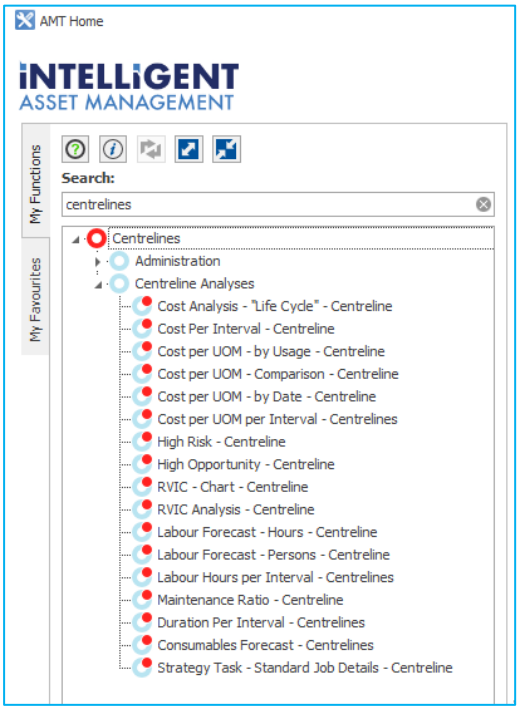
After reviewing the centreline, you can update it if you want, such as editing the data, adding new data, recalculating, and then rerun the reports.

Procedure – Analyse Centrelines

Perform the following steps to analyse centrelines:

1. To run the analysis reports:

    - Select **Centrelines** from the **My Functions** hierarchy.
    - Select **Centreline Analyses**.
    - Select the report you want from the list.



For more information on running reports, refer to the **Strategy Optimisation Training Manual.**

- *When running the reports, set the **Projection Type** as **Centreline**.*
- *As with modelling equipment, a centreline will not have any actual costs.*
- *While the steps to run centreline analysis reports are like those for running other reports in AMT, the only difference between these reports is that the centreline analysis reports will have the **- Centreline** suffix attached to them.*

Learning Check



Check your understanding of the workshop and ask any related questions.

Notes

11. Validate Life Cycle Costs

Procedure – Validate Life Cycle Costs

After all of the strategy tasks are loaded into the centreline's projection, you should validate the costs of that projection against the source data. The following reports will help this process:

- Projection Summary - Centreline
- Projection Details - Centreline
- Cost Analysis "Life Cycle" – Centreline
- Cost per UOM per Interval – Centreline
- Cost per Interval – Centreline (example below)
- Cost per UOM per Interval – Centreline

Cost Per Interval

Hierarchy : [Equipment: GS-2668 Std App: GS-2668RT] [Projection Type *: Centreline]
Analyse by : Sub System
Additional Filters : [Cost Bearer: Contract] [P/L/M:All]

	0 - 6000	6000 - 12000	12000 - 18000	18000 - 24000	Total
Air Induction & Exhaust	5,790.00	6,829.00	5,790.00	6,829.00	25,238
Brake System	0.00	4,224.00	0.00	4,224.00	8,448
Cab/Operator Station	4,309.00	4,309.00	4,309.00	4,309.00	17,236
Cooling System	0.00	25,851.00	0.00	33,099.00	58,950
Drive Train	0.00	105,772.00	0.00	105,772.00	211,544
Electrical System	7,800.00	8,713.00	8,713.00	8,713.00	33,939
Engine	9,888.00	106,711.00	9,888.00	106,711.00	233,198
Fuel System	11,355.00	0.00	11,355.00	0.00	22,710
GET	0.00	0.00	0.00	0.00	0
Hydraulic Cylinders	0.00	26,456.00	26,456.00	26,456.00	79,368
Hydraulic Pumps and Motors	0.00	28,873.00	12,945.00	28,873.00	70,691
Hydraulic System	8,352.00	27,953.00	8,352.00	27,953.00	72,610
Implements	0.00	0.00	0.00	0.00	0
Other Services	2,534.00	2,534.00	2,534.00	2,534.00	10,136
PM	26,018.18	23,187.18	23,187.18	23,187.18	95,580
Power Train	15,805.00	117,477.00	15,805.00	117,477.00	266,564
Protection & Mgmt Systems	673.00	673.00	673.00	673.00	2,692
Starting System	3,766.00	0.00	3,766.00	0.00	7,532
Steering System	0.00	26,582.00	0.00	26,582.00	53,164
Structures (Frame - Chassis)	0.00	0.00	21,231.00	0.00	21,231
Total	96,290.18	516,144.18	155,004.18	523,392.18	1,290,831

After the baseline is established, the manager may want to create a more detailed, customised centreline, perhaps to be specific to a particular application or refine how components' lives are measured or forecasted.

12. Validate and Publish

Procedure – Validate and Publish

The last step in the process is for the central management group to have the field or operations group validate the centreline. The following information should be validated:

- Equipment structure: have all the strategy tasks been identified?
- Equipment and component life cycle assumptions.
- Usage assumptions.
- Cost assumptions.
- Standard jobs and component structures.

After the relevant stakeholders approve it, the centreline should be published.

Learning Check



Check your understanding of the workshop and ask any related questions.

Notes

13. Summary

Summary

The key points to remember from this guide are:

- The create and maintain centrelines process is used to develop the desired centrelines in AMT.
- Obtain life cycle costings from OEM and from the field.
- The create and maintain centrelines process steps in AMT are:
    - Validate, edit or create centreline equipment hierarchy.
    - Update system tables (if required).
    - Create centreline equipment in AMT.
        - Create a centreline from an existing equipment.
        - Create a new blank centreline.
    - Define usage assumptions.
    - Create strategy tasks to a centreline.
    - Assign centreline standard jobs.
    - Create AMT unassigned tasks.
    - Link component structures.
    - Analyse centrelines.
    - Validate life cycle costs.
    - Validate and publish.

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